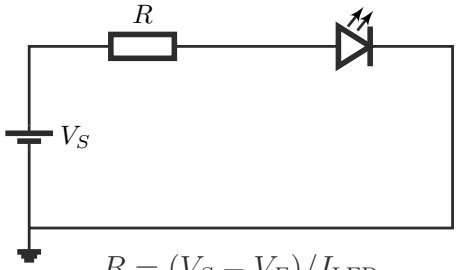
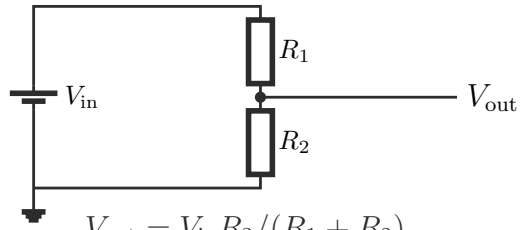
LED current limiter



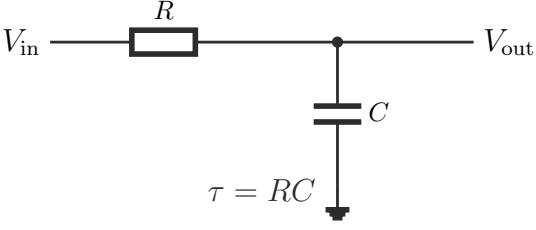
$$R = (V_S - V_F)/I_{\mathrm{LED}}$$

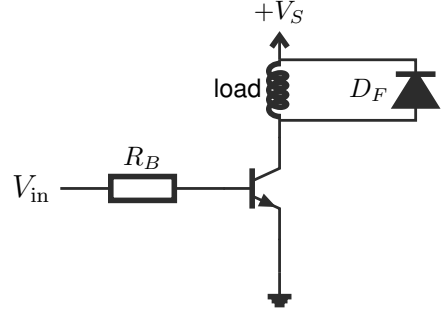
Voltage divider



$$V_{\mathrm{out}} = V_{\mathrm{in}} R_2/(R_1 + R_2)$$

RC low-pass



$$\tau = RC$$

NPN low-side driver